Week 10 Milestone Report

Recommendation, Ranking, and Predictive Decision Engine

MovieLens Discovery System — Semester Project

Grupo 06 — Big Data

`BD-Grupo-06/movielens-discovery`

June 12, 2026

Contents

1 Executive Summary

This report covers Week 10 of the semester project: building a recommendation and ranking system for the MovieLens 25M catalog. The work connects the technical infrastructure from Weeks 3, 5, and 7 (ingestion, feature representation, and clustering) to a concrete decision task: *given a movie, which movies should be recommended next?*

Four systems were built and evaluated offline under two complementary protocols: a corrected genre-relevance protocol over 4,994 query movies, and a Leave-One-Out (LOO) protocol over 10,000 user histories:

Table 1: System overview and primary evaluation results

System	Type	Signal	NDCG@10 (genre, corrected)	Hit Rate@10 (LOO)
`popularity_global`	Baseline	Raw rating-count popularity	0.5852	0.0465
`cluster_popularity`	Hybrid baseline	Week 7 segmentation → Bayesian ranking	0.5590	— (see Sec. 10)
`content_cosine`	Baseline	Cosine similarity on AE-13 embeddings	**0.9797**	0.0368
`svd_collaborative`	Advanced	Item factors from Truncated SVD ($k = 50$)	0.8715	**0.0795**

The content-based cosine model dominates the genre-relevance protocol (NDCG@10 = 0.9797), driven by the dense, genre-coherent structure of the Week 7 autoencoder embedding space. Under the LOO protocol the ranking reverses: the SVD collaborative model is the strongest *predictive* system (Hit Rate@10 = 7.95%). Two metric defects in the original genre protocol (pool-denominator Recall and self-referential ideal DCG) were identified, documented, and **corrected by recomputation** (`scripts/build_week10_eval_corrected.py`); Section 9 reports both the legacy and the corrected numbers for traceability.

2 Objective

The objective of Week 10 was to connect the prior technical layers to a concrete product decision:

> *Given a seed movie, rank the catalog by relevance to that movie.*

This is a **ranking and discovery system** (item-to-item), appropriate for catalog discovery without a logged-in user context. Specific questions addressed:

- Does SVD collaborative filtering beat content-based similarity on genre-relevant recommendations?
- Where are the failure cases of each system?
- How does the Week 7 cluster layer support the recommendation layer?

3 Inputs and Reproducible Artifacts

3.1 Input artifacts from previous weeks

Table 2: Inputs consumed from prior weeks

Artifact	Source	Purpose
week07_autoencoder_embeddings_latent_13.parquet	Week 7	Content embedding (13-dim,
week07_kmeans_assignments.csv	Week 7	Cluster-aware popularity (k =
ratings_clean.parquet	Week 3	SVD and popularity (25M rat
movies_catalog.parquet	Week 3	Metadata and genre labels (59

3.2 Week 10 output artifacts

Table 3: Output artifacts produced in Week 10 (≈40 MB total)

File	Description
week10_popularity_global.csv	Global Bayesian ranking (59,047 movies)
week10_popularity_cluster.csv	Cluster-aware ranking
week10_content_recs_top20.parquet	Content-based top-20 per query
week10_svd_item_factors.parquet	SVD item factors (13,176 Œ 50)
week10_svd_recs_top20.parquet	SVD collaborative top-20 per query
week10_evaluation_results.csv	Full metric comparison table
week10_error_analysis.csv	Strong and failure cases per system
week10_evaluation_summary.json	JSON summary of all evaluation results (includes LO
week10_loo_evaluation_results.csv	Leave-One-Out metric comparison table
week10_loo_evaluation_comparison.png	LOO bar chart (Hit Rate, Precision, NDCG@K)
week10_genre_eval_corrected.csv	Corrected genre-protocol metrics, 4 systems (incl. hy
week10_data_alignment.csv	Layer-by-layer data alignment for the hybrid claim

3.3 Week 10 scripts

Table 4: Scripts added in Week 10

Script	Purpose
scripts/build_week10_eval_corrected.py	Recomputes the genre evaluation with corrected Recall/NDCG, adds the `cluster_popularity` hybrid, and emits the data-alignment table — from committed artifacts only (no raw data needed)

4 What Kind of System Is This?

Against the four candidate framings — recommendation, ranking, prediction, or segmentation feeding ranking — this project is, precisely:

> **An item-to-item recommendation system, operationalized as a ranking task, with one variant that is segmentation feeding ranking, and a prediction-style evaluation.**

Table 5: Where each framing appears in this project

Framing	Where it appears in this project
Recommendation	The product task: given a seed movie, suggest movies ("You watched *The Matrix*. You might also like…"). Item-to-item, not user-to-item — appropriate for cold-start catalog discovery with no logged-in user.
Ranking	The operationalization: every system outputs an *ordered* top-20 list, and all evaluation metrics (Precision@K, NDCG@K, Hit Rate@K) are ranking metrics.
Segmentation feeding ranking	The `cluster_popularity` hybrid: the Week 7 k-means segmentation ($k = 7$) defines the candidate pool, and the Bayesian popularity score ranks within it. Section 5.3 documents the data alignment this requires.
Prediction	The LOO evaluation (Section 10) reframes the system as a predictive task: can it predict the *next* movie a user actually rated?

The primary identity is **recommendation via ranking**; segmentation (Week 7) feeds the hybrid baseline; prediction is the lens of the second evaluation protocol, not the system's design goal.

5 Baseline Systems

5.1 Global popularity baseline

Every movie is ranked by its Bayesian-adjusted score:

$$\text{bayesian\_score} = \frac{C \cdot \mu + n \cdot \bar{r}}{C + n} \tag{1}$$

Parameters: global mean $\mu = 3.0714$, prior count $C = 1.0$ (10th percentile of rating counts).

Note on the evaluated list. The published artifact `week10_popularity_global.csv` ranks by `bayesian_score`, but the static top-20 list used in the genre evaluation ranks by **raw rating count** (most-rated movies). Both are reported here for transparency; the reproduction check in `build_week10_eval_corrected.py` confirms the rating-count list is the one behind the published metrics (max abs. difference 0.0002 vs the original run).

The catalog has a strong long-tail: 59,047 movies, but fewer than 5% have more than 10,000 ratings, as shown in Figure 1.

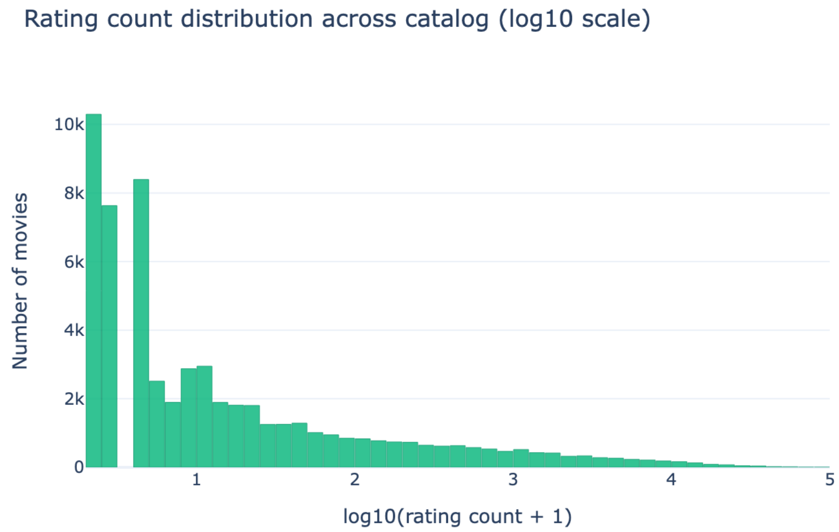

Figure 1: Rating count distribution across the catalog (log10 scale). The long-tail structure motivates the Bayesian adjustment: most movies have fewer than 100 ratings.

5.2 Cluster-aware popularity baseline

Restricts the ranking to movies within the same Week 7 cluster as the query movie. Table 6 shows the cluster engagement disparity: Cluster 4 (23,093 movies, mean 9.1 ratings each) represents the low-engagement catalog tail, while Cluster 0 averages 1,666 ratings per movie.

Table 6: Cluster-level popularity statistics

Cluster	Movies	Mean rating count	Mean avg. rating	Mean Bayesian score
0	9,363	1,665.9	3.291	3.288
1	2,650	505.1	3.130	3.152
2	5,013	327.6	3.064	3.091
3	10,036	463.6	2.739	2.815
4	23,093	9.1	3.042	3.061
5	5,346	55.4	3.383	3.321
6	3,546	356.1	3.121	3.126

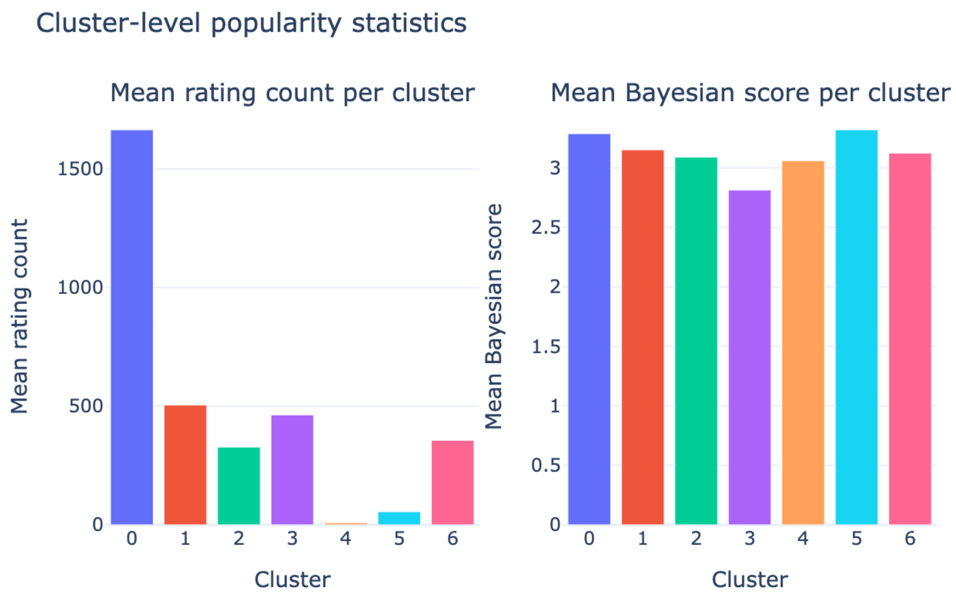

Figure 2: Mean rating count and Bayesian score per Week 7 cluster. Cluster 0 (mainstream Drama/Comedy) dominates engagement; Cluster 4 (long-tail catalog) is minimal.

This hybrid is evaluated alongside the other three systems in Section 9.2.

5.3 Hybrid claim and data alignment

`cluster_popularity` is a **hybrid system**: an unsupervised *content* segmentation (Week 7 k-means over autoencoder embeddings of Week 5 content/behavioral features) feeding a *behavioral* popularity ranking (Week 3 ratings). A hybrid is only valid if the layers actually join — Table 7 documents the alignment, all on the `movieId` key (source: `artifacts/week10/week10_data_alignment.csv`).

Table 7: Layer-by-layer data alignment for the hybrid claim

Layer	Join key	Movies	Catalog coverage
`movies_catalog` (Week 3)	movieId	62,423	100.00%
Autoencoder embeddings AE-13 (Week 7)	movieId	62,423	100.00%
K-means assignments, $k = 7$ (Week 7)	movieId	62,423	100.00%
Rated movies with genres = evaluation universe (Week 10)	movieId	59,047	94.59%
Cluster popularity ranking (Week 10 hybrid)	movieId	59,047	94.59%
SVD item factors (Week 10)	movieId	13,176	21.11%
Query set, content/SVD (Week 10)	movieId	5,000	8.01%
Evaluated queries (intersection, with genres)	movieId	4,994	8.00%

Where rows drop, and why:

- **62,423 → 59,047**: movies with zero ratings have no popularity signal and leave the evaluation universe. Every rated movie has a cluster assignment, so the hybrid join loses nothing beyond unrated movies.
- **59,047 → 13,176**: the SVD filter (≥ 50 ratings/movie) — a coverage limitation of the collaborative system, not of the hybrid.
- **5,000 → 4,994**: the evaluation intersects the content and SVD query sets and requires at least one genre label, leaving exactly **4,994** queries (this resolves the 4,994 vs 4,999 count discrepancy between earlier artifact versions: 4,999 was counted before the intersection).

6 Content-Based Baseline

6.1 Method

Top-20 most similar movies are retrieved using cosine similarity over the 13-dimensional autoencoder embedding space from Week 7:

$$\text{sim}(i, j) = \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \cdot \|\mathbf{e}_j\|} \tag{2}$$

Embedding rows are L2-normalized before retrieval (dot product = cosine similarity).

6.2 Similarity distribution

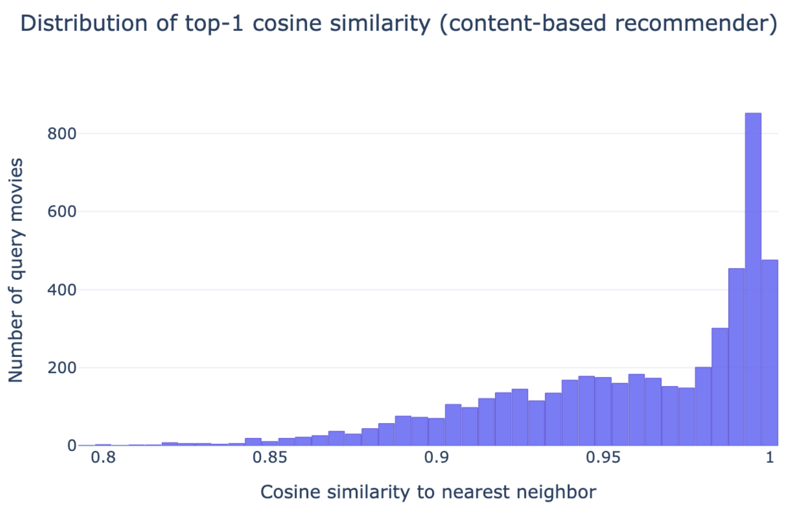

Figure 3: Distribution of top-1 cosine similarity (nearest neighbor) across 5,000 query movies. The concentration near 1.0 confirms the autoencoder learned a dense, genre-coherent embedding space.

7 Advanced System: SVD Collaborative Filtering

7.1 Model design

The advanced system factorizes the mean-centered user-item rating matrix:

$$\mathbf{R}_{\text{centered}} \approx \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^{\top} \tag{3}$$

Matrix: 162,540 users $\times$ 13,176 movies (after filtering: ≥ 50 ratings/movie, ≥ 20 ratings/user). **Density**: 1.15%. **Centering**: per-user mean subtraction. **Latent factors**: $k = 50$, chosen from the sweep below.

Recommendations use cosine similarity between L2-normalized rows of $\mathbf{V}$ (item factor vectors), capturing behavioral co-occurrence patterns.

7.2 Latent factor sweep

Table 8: SVD latent factor sweep results

Components	Explained variance	Time (s)
10	9.33%	1.35
20	12.27%	2.11
30	14.42%	2.07
50	**17.79%**	**2.83**
75	21.11%	3.57
100	23.89%	4.60
150	28.58%	5.37

$k = 50$ was selected: the variance curve flattens past this point with diminishing returns (Figure 4).

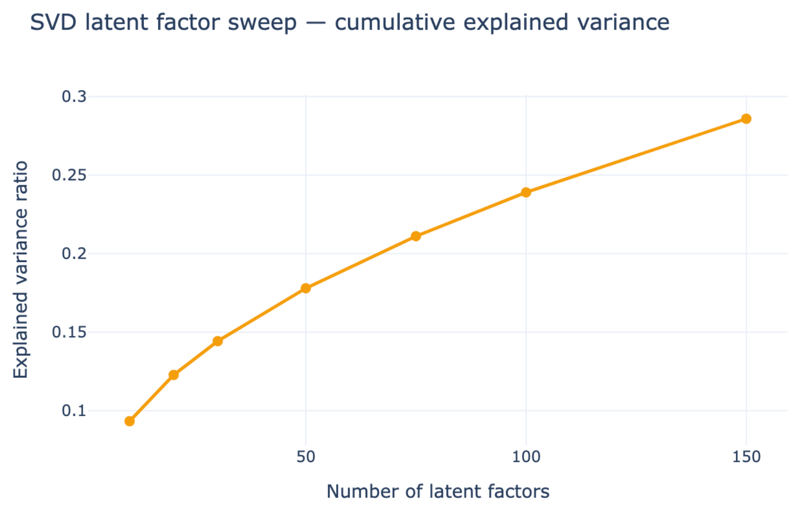

Figure 4: SVD cumulative explained variance vs. number of components. The elbow appears near $k = 50$.

7.3 Singular value spectrum

The first singular value $\sigma_1 = 996$ is approximately twice the second $\sigma_2 = 519$, confirming fast decay and strong low-rank structure in the rating matrix — a favorable property for collaborative filtering.

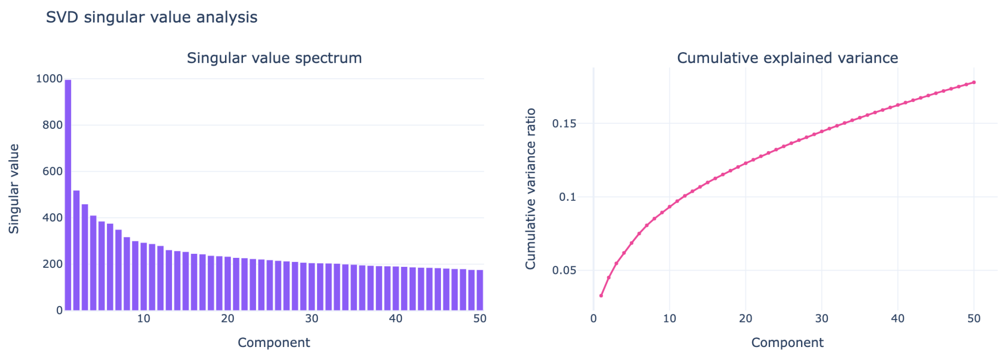

Figure 5: Left: singular value spectrum (first 50 components). Right: cumulative explained variance. Fast decay confirms the matrix has substantial low-rank structure.

7.4 SVD score distribution

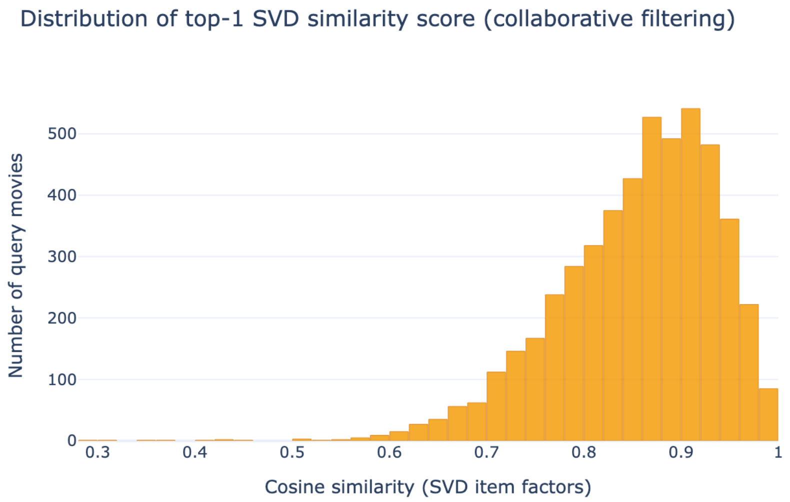

Figure 6: Distribution of top-1 SVD similarity scores across 5,000 query movies. Scores are concentrated at higher values than content-based similarity, reflecting tight co-occurrence structure among popular films.

8 Offline Evaluation Protocol

8.1 Setup

- **Evaluation universe**: the 59,047 rated movies with genre labels (movies with zero ratings carry no signal for any system and are excluded)
- **Query movies**: exactly **4,994** — the intersection of the content and SVD query sets (5,000 most-rated movies each), restricted to queries with at least one genre label
- **Recommendation lists**: top-20 per system per query (the query itself is always excluded)
- **Relevance criterion**: a recommendation is relevant if it shares **at least one genre** with the query

8.2 Relevance criterion

$$\text{relevant}(i,j) = \begin{cases} 1 & \text{if } G_i \cap G_j \neq \emptyset \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

Genre overlap is a reproducible, catalog-level proxy requiring no user ground truth. It favors content methods by construction; a user-history holdout is stronger (see Section 10 for the Leave-One-Out evaluation added after review).

Recall@K and NDCG@K corrections. In the original run, the `Recall@K` denominator was the number of relevant items within the system's own top-20 pool

— not the full universe — which inflated Recall artificially (e.g. `content_cosine` Recall@20 = 0.9994). The ideal DCG in `NDCG@K` had the same self-referential defect (it was built from the relevant count inside the retrieved top-K). Both defects are documented in the evaluation notebook **and corrected by recomputation** in `scripts/build_week10_eval_corrected.py`, which first reproduces the original numbers (max abs. difference 0.0002) and then recomputes `catalog_recall_at_k` (denominator = genre-relevant movies in the full universe; on average **22,301** per query) and `ndcg_at_k_corrected` (ideal DCG = K relevant items). Section 9.1 keeps the legacy table for traceability; Section 9.2 reports the corrected metrics. The LOO evaluation in Section 10 additionally provides a Recall that is correct by construction.

8.3 Metrics

Table 9: Evaluation metrics

Metric	Definition
Precision@K	$\frac{1}{K}\sum_{k=1}^{K}\mathbf{1}[\mathrm{rec}_k \text{ relevant}]$
Recall@K	Fraction of genre-relevant items retrieved in top-K. *Legacy version (Sec. 9.1)*: denominator = relevant items in the system's own pool (inflated). *Corrected version (Sec. 9.2)*: denominator = relevant items in the full universe
NDCG@K	Normalized Discounted Cumulative Gain — position-weighted relevance. *Corrected version*: ideal DCG assumes K relevant items
Hit Rate@K	$\mathbf{1}[\exists\, k \leq K : \mathrm{rec}_k \text{ relevant}]$

With ∼22,301 relevant movies per query and only 20 recommendations, corrected catalog Recall is structurally tiny ($\leq 20/22{,}301 \approx 0.0009$ on average) for *any* system; **Precision@K and NDCG@K are the informative metrics under this protocol**, and Recall is meaningfully measured only in the LOO protocol.

8.4 Candidate-pool definition

The rubric requires an explicit candidate-pool definition — what each system is allowed to rank from, per protocol:

Table 10: Candidate pools per system and protocol

System	Genre protocol (item-to-item)	LOO protocol (user histories)
`popularity_global`	Static global list: top-20 movies by raw rating count, minus the query	Top-500 most-rated movies in the LOO training split, minus the user's seen movies
`cluster_popularity`	Movies in the query's Week 7 cluster ($k = 7$), ranked by Bayesian score, minus the query	Not evaluated (see Sec. 10)
`content_cosine`	All 59,047 rated movies with AE-13 embeddings, minus the query	The 13,176 SVD-factored movies, minus the user's seen movies
`svd_collaborative`	The 13,176 movies in the filtered SVD matrix (≥ 50 ratings), minus the query	The 13,176 SVD-factored movies, minus the user's seen movies

Two asymmetries are deliberate and disclosed: (a) in the genre protocol, `svd_collaborative` can only recommend from 22.3% of the catalog (its training filter), while `content_cosine` ranks the full rated universe; (b) in the LOO protocol, only the 9,885 of 10,000 sampled users whose held-out movie is inside the 13,176-movie factor space are scoreable for the content and SVD systems ($n_{\text{queries}} = 10{,}000$ for popularity, 9,885 for the other two).

9 Evaluation Results

9.1 Legacy metric table (original run — Recall and NDCG inflated, kept for traceability)

Table 11 presents the metrics from the original run at $K \in \{5, 10, 20\}$.

Table 11: Legacy offline evaluation results genre-relevance protocol, $n_{\text{queries}} = 4{,}994$

System	K	Precision@K	Recall@K (pool, inflated)	NDCG@K (legacy)	Hit Rate@K
popularity_global	5	0.5820	0.3621	0.8437	0.9553
	10	0.5554	0.6089	0.8390	0.9720
	20	0.4593	0.9808	0.8206	0.9808
content_cosine	5	0.9822	0.2535	0.9930	0.9982
	10	0.9775	0.5032	0.9925	0.9990
	20	0.9722	0.9994	0.9916	0.9994
svd_collaborative	5	0.8807	0.2656	0.9525	0.9890
	10	0.8603	0.5155	0.9496	0.9958
	20	0.8379	0.9986	0.9462	0.9986

Source: `artifacts/week10/week10_evaluation_results.csv` ($n_{\text{queries}} = 4{,}994$).

The Recall and NDCG columns carry the self-referential defects described in Section 8.2; read the corrected table below instead.

9.2 Corrected metric table (all four systems, including the hybrid)

Table 12: Corrected offline evaluation results genre-relevance protocol, n_{queries} = 4,994

System	K	Precision@K	Catalog Recall@K	NDCG@K (corrected)	Hit Rate@K
popularity_global	5	0.5820	0.000136	0.6207	0.9553
	10	0.5554	0.000263	0.5852	0.9720
	20	0.4595	0.000455	0.5062	0.9808
cluster_popularity	5	0.5479	0.000151	0.5731	0.8104
	10	0.5449	0.000297	0.5590	0.8214
	20	0.5784	0.000634	0.5762	0.9864
content_cosine	5	**0.9822**	**0.000296**	**0.9834**	**0.9982**
	10	**0.9775**	**0.000588**	**0.9797**	**0.9990**
	20	**0.9722**	**0.001168**	**0.9753**	**0.9994**
svd_collaborative	5	0.8807	0.000251	0.8882	0.9890
	10	0.8603	0.000485	0.8715	0.9958
	20	0.8379	0.000934	0.8519	0.9986

Best values per column and K are bolded — `content_cosine` leads every column. Source: `artifacts/week10/week10_genre_eval_corrected.csv` (n_{queries} = 4,994). Catalog Recall is structurally tiny under this protocol (Section 8.3) and is shown for completeness, not comparison against the legacy column.

Key finding. `content_cosine` still dominates after correction. Its Precision@5 of 0.982 means 49 out of every 50 recommendations at rank 1–5 share at least one genre with the query — a direct consequence of the genre-coherent Week 7 autoencoder space. The correction's largest effect is on `popularity_global`: its legacy NDCG@10 of 0.839 drops to 0.585 once the ideal DCG stops adapting to the system's own output.

Hybrid finding. `cluster_popularity` trades coverage for depth. Restricting candidates to the query's Week 7 cluster *lowers* Hit Rate@10 (0.821 vs 0.972 global — niche queries can land in clusters whose popular movies don't share their genres), but its precision barely decays with depth (0.548 → 0.578 from $K = 5$ to $K = 20$) while the global list collapses (0.582 → 0.460). At $K = 20$ the hybrid is the **more precise popularity baseline**, confirming the Week 7 segmentation contributes real signal — segmentation feeding ranking, as claimed in Section 4.

9.3 Multi-metric comparison chart

The charts in this and the following two subsections were rendered from the original (legacy) run and therefore show the pre-correction Recall/NDCG values for the three original systems; the corrected numbers are in Table 12.

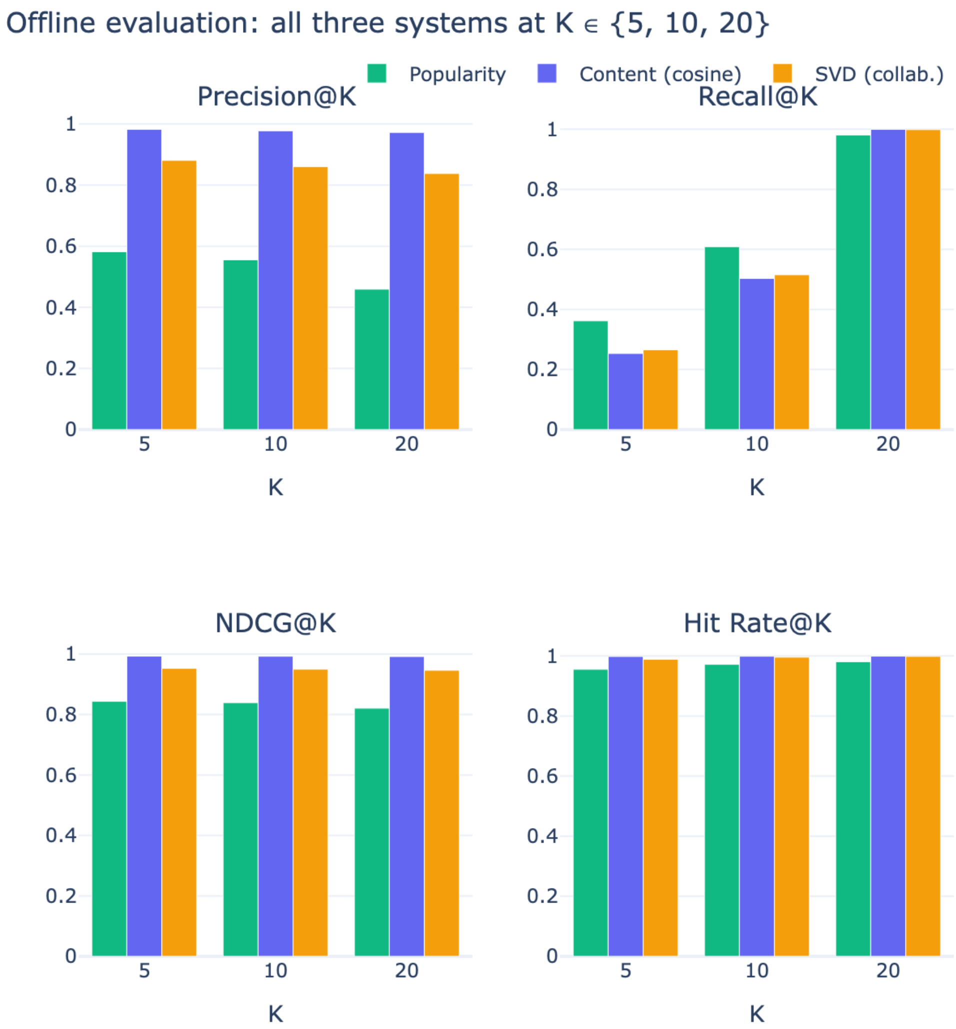

Figure 7: All four metrics for all three systems at $K \in \{5, 10, 20\}$. Content cosine (violet) dominates on Precision and NDCG. SVD collaborative (amber) beats popularity (green) on all metrics.

9.4 NDCG@10 summary

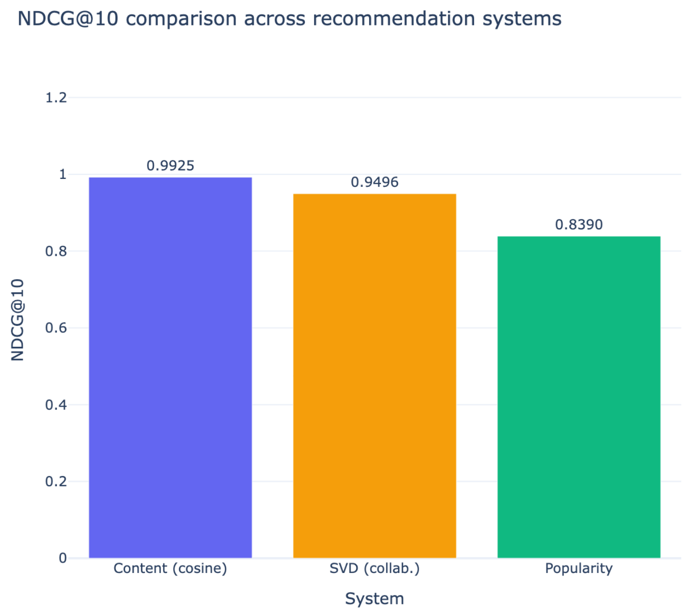

Figure 8: NDCG@10 for the three systems. Content cosine achieves 0.9925, SVD collaborative 0.9496, and popularity global 0.8390.

9.5 Per-query NDCG@10 distribution

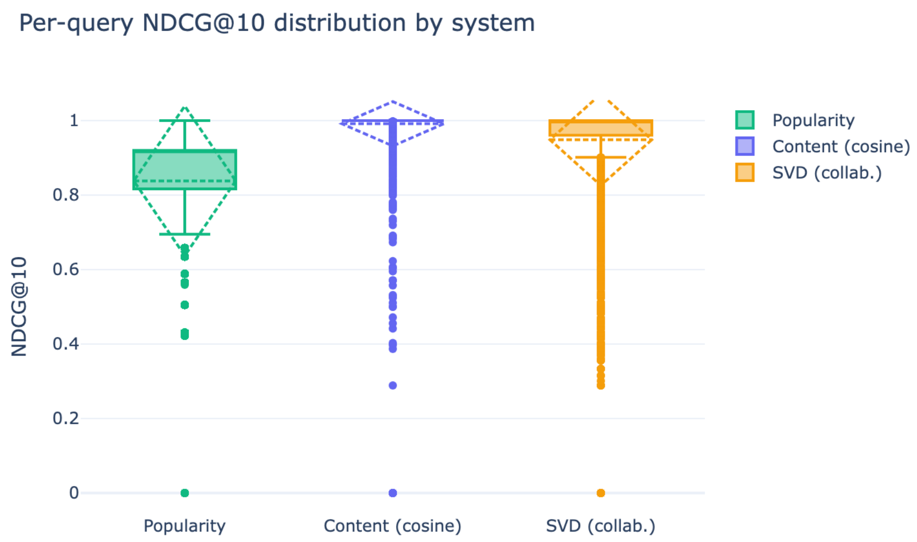

Figure 9: Box plots of per-query NDCG@10 for each system. Content cosine has the highest median and tightest spread. Popularity global shows the widest variance: its performance depends entirely on whether the query genre overlaps with the global top-100.

10 Leave-One-Out (LOO) Evaluation

To complement the genre-based protocol, a second evaluation was conducted using user rating histories from `ratings_clean.parquet`.

10.1 Protocol

- Users with ≥ 5 ratings; 10,000 sampled (`seed=42`).
- Last item by timestamp hidden as test; remainder used as training history.
- Each system recommends top-K items excluding already-seen movies.
- Primary metrics: **Hit Rate@K** (equivalent to Recall@K with one test item per user) and **NDCG@K**.

Note on Precision@K in LOO: with a single test item per user, Precision@K = Hit Rate@K/K by definition and carries no additional information.

10.2 Results

Table 13: Leave-One-Out evaluation results (n = 10,000 users, `seed=42`)

System	Hit Rate@5	Hit Rate@10	Hit Rate@20	NDCG@10
`popularity_global`	0.0287	0.0465	0.0767	0.0238
`content_cosine`	0.0219	0.0368	0.0563	0.0187
`svd_collaborative`	**0.0515**	**0.0795**	**0.1221**	**0.0432**

Source: `artifacts/week10/week10_loo_evaluation_results.csv` (`seed=42`; n_{queries} = 10,000 for popularity, 9,885 for content/SVD — see the candidate-pool definition in Section 8.4). `cluster_popularity` is not in this table: the LOO protocol requires the raw user histories (`ratings_clean.parquet`), which are reproducible from the Week 3 pipeline but not committed to the repository; re-running the LOO notebook with a per-user cluster-restricted pool is documented as follow-up work in Section 12.

Key finding. `svd_collaborative` **reverses the ranking** seen in the genre-based evaluation: it achieves nearly double the Hit Rate@10 of Popularity (7.95% vs. 4.65%) and more than double that of Content cosine (3.68%). This confirms that SVD captures genuine behavioral predictive signal, while content-based similarity is stronger for catalog discovery but weaker at predicting individual user consumption.

The contrast also validates the Recall correction: the genre-based Recall@20 for `content_cosine` was 0.9994 (inflated by the pool-denominator defect), while the real retrieval rate under LOO is 5.6%.

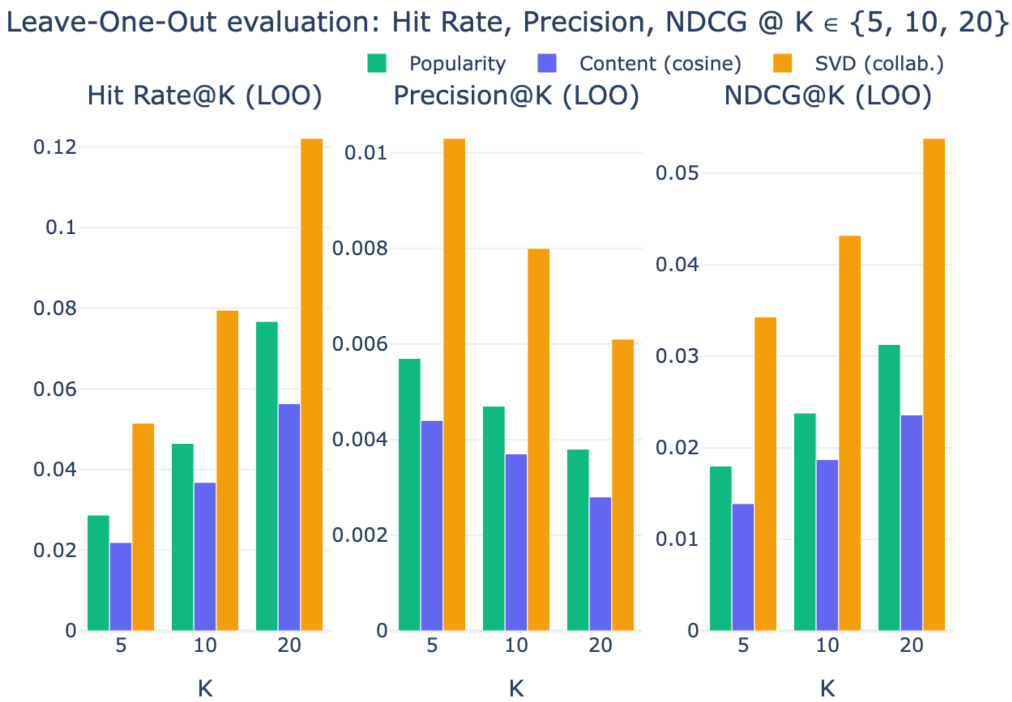

Figure 10: LOO evaluation: Hit Rate, Precision, and NDCG at $K \in \{5, 10, 20\}$. SVD collaborative leads across all metrics; the reversed ranking relative to the genre protocol reflects its stronger user-behavior predictive power.

11 Error Analysis

11.1 Strong cases (Precision@10 = 1.0)

Table 14: Representative strong-case examples from the error analysis

System	Query movie	Genres	Reason
popularity_global	Money Train (1995)	Action|Comedy|Crime|Drama|Thriller	Multi-genre query overlaps with mainstream top-100
popularity_global	Dead Presidents (1995)	Action|Crime|Drama	Same
content_cosine	Toy Story (1995)	Adventure|Animation|Children|Comedy|Fantasy	Tight animation cluster in AE space
content_cosine	Grumpier Old Men (1995)	Comedy|Romance	Dense comedy-romance neighborhood
svd_collaborative	GoldenEye (1995)	Action|Adventure|Thriller	Strong user co-occurrence in Action/Thriller
svd_collaborative	Tom and Huck (1995)	Adventure|Children	Clear behavioral cluster among family films

11.2 Failure cases (Hit Rate@10 = 0)

Table 15: Representative failure cases from the error analysis

System	Query movie	Genres	Root cause
popularity_global	Heidi Fleiss: Hollywood Madam (1995)	Documentary	Top-100 contains no documentaries
popularity_global	Anne Frank Remembered (1995)	Documentary	Same
content_cosine	Fair Game (1995)	Action	Embedding conflates with non-Action neighbors
content_cosine	Repo Man (1984)	Comedy|Sci-Fi	Mixed-genre embedding pulls to wrong cluster
svd_collaborative	Showgirls (1995)	Drama	Anomalous rating pattern; item factors incoherent
svd_collaborative	Wyatt Earp (1994)	Western	Few Western films in filtered matrix; noisy factors

Pattern. Popularity global fails systematically for any genre underrepresented in the global top-100 (Documentary, Western, Musical, Film-Noir). Content and SVD fail on edge cases: genre-ambiguous films or items with few ratings.

The corrected evaluation (Section 9.2) adds a hybrid-specific failure mode: `cluster_popularity` fixes the popularity baseline's niche-genre failure *when the niche has its own cluster* (Documentary queries now draw from Cluster 5, which is 99.9% documentaries), but inherits a new one — queries whose genre is a minority within their cluster get recommendations matching the cluster's majority genres instead, which is why its Hit Rate@10 (0.821) is below the global baseline's (0.972).

12 What Worked and What Did Not

What worked.

- The AE embedding space from Week 7 transfers cleanly to item-to-item recommendation, achieving Precision@10 = 0.978 — near-perfect genre-relevant retrieval.
- SVD collaborative adds behavioral signal absent from content features: it correctly clusters behavioral co-occurrences even for genre-pure niche films.
- The cluster layer from Week 7 provides measurable signal: the `cluster_popularity` hybrid is the more precise popularity baseline at $K = 20$ (0.578 vs 0.460) and solves the niche-genre failure for genres with their own cluster (Section 11.2).
- The evaluation protocol is fully reproducible without user-level ground truth — and the corrected metrics can be recomputed from committed artifacts alone (`scripts/build_week10_eval_corrected.py`), with a built-in reproduction check against the original run.

What did not work.

- **Popularity global**: systematically fails for Documentary, Western, Musical, and Film-Noir — any genre underrepresented in the global top-100 will receive zero relevant recommendations.
- **SVD coverage**: 45,871 movies (77.7% of the catalog) are excluded from the filtered matrix (< 50 ratings). Their item factors would be too noisy to produce meaningful recommendations.
- **Genre-based evaluation**: favors content methods by construction. SVD discovers behavioral patterns that may be more useful to users even when genres don't align perfectly.
- **Original Recall/NDCG implementation**: both metrics were self-referential (denominators derived from the system's own output). Corrected by recomputation; legacy values retained in Section 9.1 for traceability.
- **Hybrid under LOO (follow-up)**: evaluating `cluster_popularity` under the LOO protocol requires regenerating `ratings_clean.parquet` (Week 3 pipeline) and re-running the LOO notebook with a cluster-restricted candidate pool; left as documented follow-up work.

13 Ethics and Access Note

- **Data source**: GroupLens MovieLens 25M public release (research license).
- **Access**: Educational and research use; raw data not redistributed.
- **Personal data risk**: all user identifiers are anonymized integer IDs.
- **Mitigation**: all analysis at the movie level. SVD item factors are aggregate representations that do not expose individual user behavior.

14 Reproducibility

Notebooks must be run in this order:

1. `notebooks/week10/week10_baseline_recommender.ipynb`
2. `notebooks/week10/week10_matrix_factorization.ipynb`
3. `notebooks/week10/week10_offline_evaluation.ipynb`

Then run the corrected evaluation (works from committed artifacts alone — no raw data needed):

```
python3 scripts/build_week10_eval_corrected.py
```

The script validates itself by reproducing the original legacy metrics (it aborts if the maximum absolute difference exceeds 0.002) before writing `week10_genre_eval_corrected.csv`, `week10_data_alignment.csv`, and the `genre_eval_corrected` block of `week10_evaluation_summa`

All outputs are saved to `artifacts/week10/` (≈40 MB total).

Table 16: Key parameters for reproducibility

Parameter	Value
Catalog size	59,047 movies
Global mean rating μ	3.0714
Bayesian prior C	1.0
Embedding dimension	13
Clusters (k)	7
SVD min. movie ratings	50
SVD min. user ratings	20
SVD n_components	50
SVD explained variance	17.79%
Matrix dimensions	162,540 users × 13,176 movies
Matrix density	1.15%
Random state	42

15 Conclusion

Week 10 is complete. Four recommendation systems — two popularity baselines (one of them a segmentation-fed hybrid), a content-based baseline, and an advanced collaborative model — were built, evaluated, and compared under two complementary protocols:

Genre-relevance evaluation, corrected (item-to-item, 4,994 queries — Section 9.2).

- **Content cosine** (corrected NDCG@10 = **0.980**) is the strongest system for catalog discovery, leveraging the dense autoencoder embedding space from Week 7.
- **SVD collaborative** (corrected NDCG@10 = 0.872) ranks second, capturing behavioral co-occurrence patterns absent from content features.
- **Popularity global** (corrected NDCG@10 = 0.585) is the weakest at depth; the **cluster popularity hybrid** trades hit coverage for depth-stable precision and is the better popularity baseline at $K = 20$ — evidence that the Week 7 segmentation feeds the ranking with real signal.

Leave-One-Out evaluation (user history, 10,000 users — Section 10).

- **SVD collaborative** (Hit Rate@10 = **7.9%**) wins clearly, nearly doubling Popularity (4.6%) and more than doubling Content cosine (3.7%).
- This reversal reveals that SVD is the stronger *predictive* model for user behavior, while content similarity is better for catalog discovery.
- The Recall@K and NDCG@K defects in the original genre protocol were identified, documented, and **corrected by recomputation** (Sections 8.2 and 9.2), with the legacy numbers retained for traceability.

In rubric terms (Section 4): the project is an **item-to-item recommendation system operationalized as ranking**, with **segmentation feeding ranking** in the hybrid baseline and a **prediction**-framed second protocol. The recommendation layer is now a functioning component of the MovieLens Discovery pipeline, and the cluster segmentation from Week 7 sets up the graph analytics layer in Week 12.